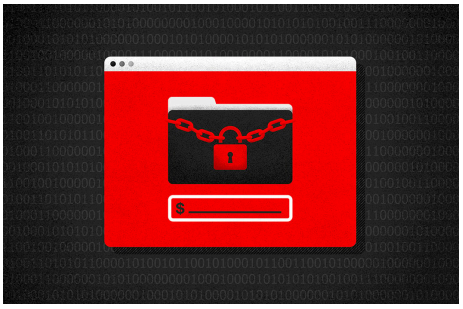
Decryptable PartyTicket Ransomware Reportedly Targeting Ukrainian Entities

March 01, 2022 | CrowdStrike Intelligence Team | Counter Adversary Operations



Summary

On Feb. 23, 2022, destructive attacks were conducted against Ukrainian entities. Industry reporting has claimed the Go-based ransomware dubbed *PartyTicket* (or *HermeticRansom*) was identified at several organizations affected by the attack,[1] among other families including a sophisticated wiper CrowdStrike Intelligence tracks as *DriveSlayer* (*HermeticWiper*).

Analysis of the *PartyTicket* ransomware indicates it superficially encrypts files and does not properly initialize the encryption key, making the encrypted file with the associated `.encryptedJB` extension recoverable.

Technical Analysis

A *PartyTicket* ransomware sample has a SHA256 hash of `4dc13bb83a16d4ff9865a51b3e4d24112327c526c1392e14d56f20d6f4eaf382`. It has been observed associated with the file names `cdir.exe`, `cname.exe`, `connh.exe` and `intpub.exe`.

The ransomware sample — written using Go version 1.10.1 — contains many symbols that reference the U.S. political system, including `voteFor403`, `C:/projects/403forBiden/wHiteHousE` and `primaryElectionProcess`.

The ransomware iterates over all drive letters and recursively enumerates the files in each drive and its subfolders, excluding file paths that contain the strings `Windows` and `Program Files` and the folder path `C:\Documents and Settings` (the latter folder was replaced in Windows versions later than Windows XP with `C:\Users`). Files with the following extensions are selected for encryption:

`acl, avi, bat, bmp, cab, cfg, chm, cmd, com, contact, crt, css, dat, dip, dll, doc, docx, dot, encryptedjb , epub, exe, gif, htm, html, ico, in, iso, jpeg, jpg, mp3, msi, odt, one, ova, pdf, pgsql, png, ppt, pptx, pub, rar, rtf, sfx, sql, txt, url, vdi, vsd, wma, wmv, wtv, xls, xlsx, xml, xps, zip`

For each file path that passes the previously described path and extension checks, the ransomware copies an instance of itself to the same directory it was executed from and executes via the command line, passing the file path as an argument. The parent ransomware process names its clones with a random UUID generated by a public library[2] that uses the current timestamp and MAC addresses of the infected host's network adapters. The malware developer attempted to use Go's `WaitGroup` types to implement concurrency; however, due to a likely coding error, the ransomware creates a very large number of threads (one per enumerated file path) and copies its own binary into the current directory as many times as there are selected files. After all encryption threads have ended, the original binary deletes itself via the command line.

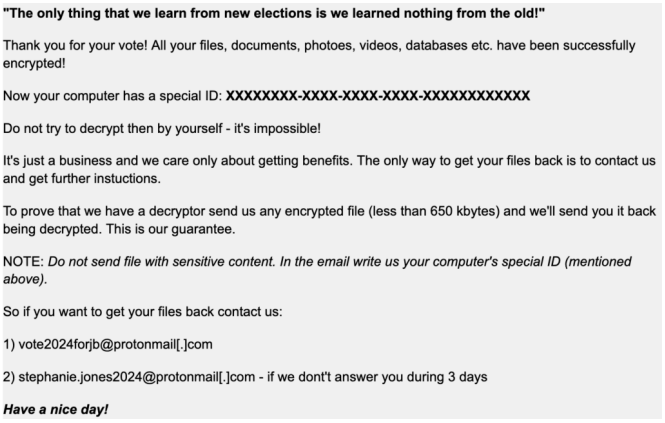
When the sample receives a file path as an argument, it encrypts the file using AES in Galois/Counter Mode (GCM). The AES key is generated using the Go `rand` package's `Intn` function to select offsets in the character array `1234567890ABCDEFGHIJKLMNOPQRSTUVWXYZ`, generating a 32-byte key. Due to another likely coding error, the seed for the `Intn` function is updated after the key is generated, meaning the same AES key is generated each time the binary and its clones are run. All of the files encrypted on a host are encrypted with the same key, and knowledge of the corresponding *PartyTicket* sample's key enables their decryption. A script using this flaw to recover the encrypted files is available on the CrowdStrike Git Repository.

For each file, the AES encryption key is itself encrypted with RSA-OAEP, using a public RSA key that has the following parameters:

```
Modulus (N):
0xcbb94cb189a638b51e7cfe161cd92edb7145ecbd93989
e78c94f8c15c61829286fd834d80c931daed4ac4aba1483
5fd3a6721602bcaa7193245fc6cf8e1d3261460ff3f1cba
e3d44690beb989adee69ac486a932ee44dbdf0a44e772ab
9822a16753cd08bdbb169f866f722114ee69c0cf1588fda
f8f7efd1c3ed243786078593f9b0cd867bab2b170c18436
60d16e2181ae679137e2650551a41631398e027206e22a5
5858c741079ceafd50d5bd69546d4d52f5a33b0a576e175
0d3f83afa1ce4403d768cbd670b443f61794b44705a8b11
32c0c0ce77dbd04053ba20aec9baf23944270f10d16ad07
27ed490c91c7f469278827c20a3e560f7c84015f7e1b
Exponent (E): 0x10001
```

Before encryption, the ransomware renames the file using the format `<original file name>.com>.encryptedJB` ("JB" very likely stands for the initials of the United States president Joseph Biden, given the other political content in the binary). The ransomware then overwrites the content with the encrypted data. *PartyTicket* will only encrypt the first 9437184 bytes (9.44 MB) of a file. If the file passed as an argument is larger than this limit, any data above it is left unencrypted. After the file contents are encrypted, *PartyTicket* appends the RSA-encrypted AES key at the end of the file.

The ransomware also writes an HTML ransom note on the user's desktop directory with the name `read_me.html` before the file encryption starts (Figure 1). Unless they are intentional mistakes, grammar constructs within the note suggest it was likely not written or proofread by a fluent English speaker.



"The only thing that we learn from new elections is we learned nothing from the old!"

Thank you for your vote! All your files, documents, photoes, videos, databases etc. have been successfully encrypted!

Now your computer has a special ID: **XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX**

Do not try to decrypt then by yourself - it's impossible!

It's just a business and we care only about getting benefits. The only way to get your files back is to contact us and get further instuctions.

To prove that we have a decryptor send us any encrypted file (less than 650 kbytes) and we'll send you it back being decrypted. This is our guarantee.

NOTE: *Do not send file with sensitive content. In the email write us your computer's special ID (mentioned above).*

So if you want to get your files back contact us:

1) vote2024forjb@protonmail[.]com

2) stephanie.jones2024@protonmail[.]com - if we dont't answer you during 3 days

Have a nice day!

Figure 1. Ransom note

Assessment

CrowdStrike Intelligence does not attribute the *PartyTicket* activity to a named adversary at the time of writing.

The ransomware contains implementation errors, making its encryption breakable and slow. This flaw suggests that the malware author was either inexperienced writing in Go or invested limited efforts in testing the malware, possibly because the available development time was limited. In particular, *PartyTicket* is not as advanced as *DriveSlayer,* which implements low-level NTFS parsing logic. The relative immaturity and political messaging of the ransomware, the deployment timing and the targeting of Ukrainian entities are consistent with its use as an additional payload alongside *DriveSlayer* activity, rather than as a legitimate ransomware extortion attempt.

YARA Signatures

The following YARA rule can be used to detect *PartyTicket*:

```
rule CrowdStrike_PartyTicket_01 : ransomware
golang
{
    meta:
        copyright = "(c) 2022 CrowdStrike Inc."
        description = "Detects Golang-based
crypter"
        version = "202202250130"
        last_modified = "2022-02-25"
    strings:
        $ = ".encryptedJB" ascii
        $start = { ff 20 47 6f 20 62 75 69 6c
64 20 49 44 3a 20 22 }
        $end = { 0a 20 ff }
    condition:
        uint16(0) == 0x5A4D and
uint32(uint32(0x3C)) == 0x00004550 and
        for 1 of ($end) : ( @start < @ and
@start + 1024 > @) and
        all of them
}

rule CrowdStrike_PartyTicket_02 : PartyTicket
golang
{
    meta:
        copyright = "(c) 2022 CrowdStrike Inc."
        description = "Detects Golang-based
PartyTicket ransomware"
        version = "202202250130"
        last_modified = "2022-02-25"
      strings:
        $s1 = "voteFor403"
        $s2 = "highWay60"
        $s3 = "randomiseDuration"
        $s4 = "subscribeNewPartyMember"
        $s5 = "primaryElectionProces"
        $s6 = "baggageGatherings"
        $s7 = "getBoo"
        $s8 = "selfElect"
        $s9 = "wHiteHousE"
        $s10 = "encryptedJB"
        $goid = { ff 20 47 6f 20 62 75 69 6c 64
20 49 44 3a 20 22 71 62 30 48 37 41 64 57 41 59
44 7a 66 4d 41 31 4a 38 30 42 2f 6e 4a 39 46 46
38 66 75 70 4a 6c 34 71 6e 45 34 57 76 41 35 2f
50 57 6b 77 45 4a 66 4b 55 72 52 62 59 4e 35 39
5f 4a 62 61 2f 32 6f 30 56 49 79 76 71 49 4e 46
62 4c 73 44 73 46 79 4c 32 22 0a 20 ff }
        $pdb =
"C://projects//403forBiden//wHiteHousE"
    condition:
        (uint32(0) == 0x464c457f or (uint16(0)
== 0x5a4d and uint16(uint32(0x3c)) == 0x4550))
```

```
and 4 of ($s*) or $pdb or $goid
}
```

Script to Decrypt *PartyTicket* Encrypted Files

Due to the previously discussed implementation errors in the AES key generation, it is possible to recover the AES key used for encryption by *PartyTicket*. The below Go script decrypts files encrypted by *PartyTicket* sample `4dc13bb83a16d4ff9865a51b3e4d24112327c526c1392e14d56f20d6f4eaf382`. The script takes the file to be decrypted as an argument via the "-p" flag and saves the decrypted output to "decrypted.bin" in the same directory. The script can be built as an executable or run via the Go run package; it was tested using Go version go1.16.6.

```
package main

import (
	"crypto/aes"
	"crypto/cipher"
	"encoding/hex"
	"fmt"
	"os"
	"flag"
)

func main() {

	encrypted_filepath := flag.String("p",
"encrypted.bin", "Path to encrypted file")
	flag.Parse()

	fmt.Printf("Decrypting file : %s\n",
*encrypted_filepath)
	key_bytes :=
<>byte("6FBBD7P95OE8UT5QRTTEBIWAR88S74DO")
	key := hex.EncodeToString(key_bytes)
	fmt.Printf("Decryption key : %s\n",
key_bytes)

	dat, err :=
os.ReadFile(*encrypted_filepath)
	if err != nil {
		fmt.Println("Unable to open file,
please supply path of encrypted file with flag
-p, default file path is ./encrypted.bin")
		os.Exit(3)
	}

	decrypted_filepath := "decrypted.bin"
	filecontents := dat
	encrypted_contents :=
filecontents<:len(filecontents) - 288>
	enc_size := len(encrypted_contents)
	bsize := 1048604
	cycles := enc_size / bsize

	if cycles == 0{

		encrypted :=
hex.EncodeToString(encrypted_contents)
		decrypted := decrypt(encrypted,
key)
		write_output(decrypted_filepath,
decrypted)
		} else {
			for i:=0; i<cycles; i++ {
if i >= 9 {
				start := 9
```

```
* bsize
                                        end :=
enc_size
                                        data :=
string(encrypted_contents)

write_output(decrypted_filepath, data)
                                        break
                                }
                                block_start := i *
bsize
                                block_end := (i+1)
* bsize
                                if block_end >
enc_size{
                                        block_end
:= enc_size

encrypted:=hex.EncodeToString(encrypted_content
s)
                                        decrypted
:= decrypt(encrypted, key)

write_output(decrypted_filepath, decrypted)

                                }


encrypted:=hex.EncodeToString(encrypted_content
s)
                                decrypted :=
decrypt(encrypted, key)

write_output(decrypted_filepath, decrypted)
                        }
                }

                fmt.Printf("Decrypted file
written to : %s\n", decrypted_filepath)

        }

func write_output(filepath string, data string)
{
                f, err := os.OpenFile(filepath,
os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
                if err != nil {
                        panic(err)
                }
                byte_data := <>byte(data)
                f.Write(byte_data)
                f.Close()
}

func decrypt(encryptedString string, keyString
```

```
string) (decryptedString string) {

	key, _ := hex.DecodeString(keyString)
	enc, _ :=
hex.DecodeString(encryptedString)

	block, err := aes.NewCipher(key)
	if err != nil {
		panic(err.Error())
	}
	aesGCM, err := cipher.NewGCM(block)
	if err != nil {
		panic(err.Error())
	}
	nonceSize := aesGCM.NonceSize()
	nonce, ciphertext := enc<:nonceSize>,
enc
	plaintext, err := aesGCM.Open(nil,
nonce, ciphertext, nil)
	if err != nil {
		panic(err.Error())
	}

	return fmt.Sprintf("%s", plaintext)
}
```

Endnotes

1. https<:>//symantec-enterprise-blogs.security<.>com/blogs/threat-intelligence/ukraine-wiper-malware-russia
2. https<:>//pkg.go.dev/github<.>com/satori/go.uuid#NewV1

Additional Resources

- *Read more about use of offensive cyber operations against Ukraine: Lessons Learned From Successive Use of Offensive Cyber Operations Against Ukraine and What May Be Next.*
- *Learn how CrowdStrike Falcon® provides continuous protection against DriveSlayer and wiper-style threats: CrowdStrike Falcon® Protects from New Wiper Malware Used in Ukraine Cyberattacks*
- *Read about WhisperGate in this CrowdStrike Intelligence blog: Technical Analysis of the WhisperGate Malicious Bootloader.*
- *Learn about the powerful, cloud-native CrowdStrike Falcon® platform by visiting the product webpage.*
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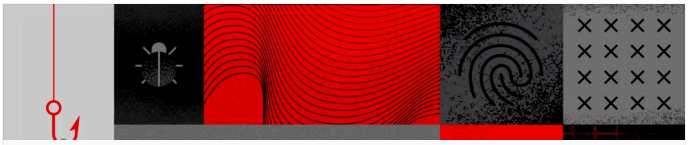

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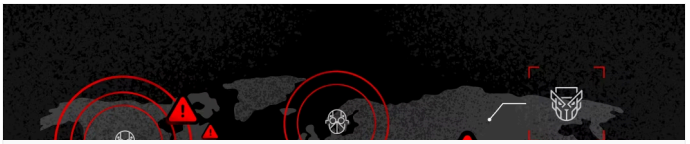

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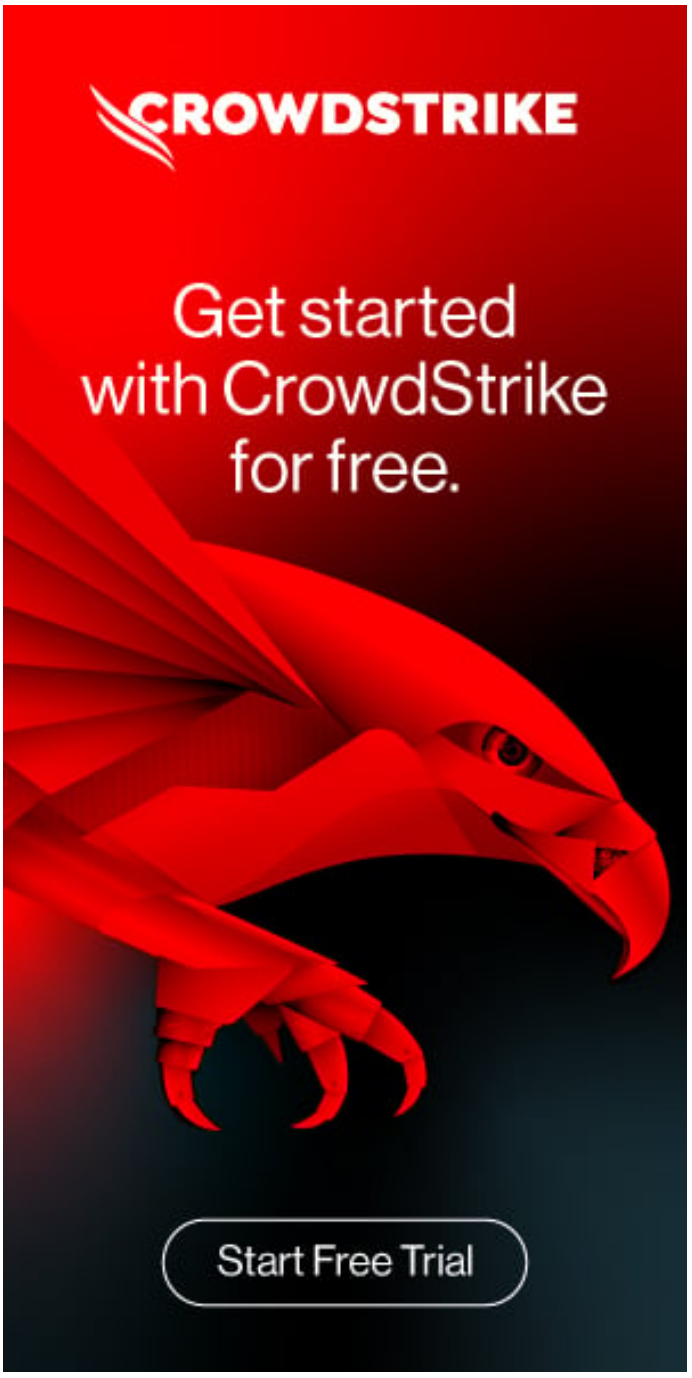

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